HeatSense

A Physics-Informed Cell-Level Digital Twin for Battery State Estimation and Thermal-Anomaly Precursor Monitoring

Final Idea Report – Technical Documentation

Project stage: ideation · Hardware implementation: not started · Results: none yet

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Executive Summary

HeatSense is a capstone project that proposes a cell-level, physics-informed Digital Twin for lithium-ion battery state estimation and for early detection of thermal and impedance anomalies. The design combines a deterministic microcontroller tier, which samples voltage, current and surface temperature and owns every safety cut-off, with a host-computer tier that runs a second-order Thevenin equivalent-circuit model (2RC ECM), an Extended Kalman Filter (EKF), a two-node thermal model and a lightweight physics-informed machine-learning layer. The machine-learning layer is advisory only: it estimates a State of Health (SoH) proxy and an anomaly score, and it has no authority over the hardware relay.

The project is at the ideation stage. No hardware has been assembled, no firmware has been written, and no measurement or accuracy figure exists yet. Every quantitative result in this document is therefore written as "[not yet measured]". The central research question is whether physics, state estimation and machine learning together outperform or usefully complement a direct sensor-to-ML model on the same data. The project is built so that this question can be answered with a falsifiable experiment rather than assumed.

> **Decisions locked for this report**
> Scope: one 18650 cell. Cell → module → pack scaling is future work and is not validated.
> Wording: "thermal and impedance precursor anomaly detection". HeatSense does not predict or prevent thermal runaway.

> SoC, SoH and RUL are estimates. “Normal” means no configured threshold was violated; it does not mean the battery is safe.
> Safety cut-off is deterministic firmware plus a relay. AI never actuates it.
> No Transformer or foundation-model claims in the main project.

1. Problem Statement

Battery management systems protect lithium-ion cells mainly by comparing externally measurable quantities (terminal voltage, current and surface temperature) with fixed limits. Protection of this kind is necessary and reliable, but it acts only when a limit is crossed. The processes that degrade a cell, such as growth of the solid electrolyte interphase (SEI) layer and the resulting rise in internal resistance, occur inside the cell. The quantities that describe them (internal resistance, polarization voltages and core temperature) cannot be read from the terminals or the case.

Many BMS implementations already estimate state of charge and state of health, so the gap is not that estimation is impossible. The open question is whether a real-time physical model, kept in step with live measurements, can reveal impedance and thermal precursors earlier than a threshold would, using hardware modest enough to run at the edge, and without weakening the deterministic safety function. HeatSense is designed to test exactly that.

1.1 Why this matters

- Cost: the battery pack is reported in the project sources to be roughly 40 to 60 percent of EV cost, so degradation that is found late is expensive to correct. This figure is source-reported and has not been independently verified for this report.
- Visibility: surface sensors lag the cell interior, so a surface reading can understate conditions inside the cell during dynamic load.
- Safety and trust: a learned model that cannot explain its output, or that can violate physical limits, is hard to place near a safety function.

2. Proposed Solution

HeatSense proposes a cell-level Digital Twin that synchronizes live voltage, current and temperature measurements with a 2RC equivalent-circuit model, an Extended Kalman Filter and a two-node thermal model. The filter estimates hidden states (state of charge, polarization voltages and internal resistance) and the thermal model estimates a core-temperature state from the surface reading and the heat generated inside the cell. A lightweight physics-informed ML layer then uses these estimated states, and the filter residuals, to produce a SoH proxy and a thermal-anomaly score.

Here “Digital Twin” has a narrow meaning: a real-time cyber-physical state estimator that keeps a parametric electrical and thermal model synchronized with one physical cell and uses it to track states that sensors cannot see. It is not a 3D visualization and it is not a pack-level simulator.

2.1 Design principles

1. **Separate safety from intelligence.** A microcontroller with fixed thresholds and a relay is the only path that can disconnect the cell.

2. **Prefer physics where physics is known.** The ECM, EKF and thermal model supply interpretable states; ML is used only for what the physics does not give directly, such as longer-horizon health trends.
3. **Make the claim falsifiable.** The ML layer is benchmarked against a direct sensor-to-ML baseline. If the Digital Twin does not help, the report says so.
4. **State what is estimated.** Every non-measured quantity is labelled as an estimate.
5. **Keep scope honest.** One cell, non-destructive testing only, no certification claims.

3. Background and Rationale

3.1 Observable and latent quantities

A cell exposes only a few quantities to the outside world. Everything else that matters for health and safety must be inferred from a model. Table 1 separates the two groups.

Directly measured (planned sensors)	Estimated / latent (Digital Twin outputs)
Terminal voltage V (INA219)	State of Charge (SoC)
Load current I (INA219)	Polarization voltages V_1, V_2
Surface temperature $T_{surface}$ (3× DS18B20)	Internal ohmic resistance R_s
Ambient temperature (optional BME280)	Core temperature T_{core} (thermal-model estimate)
	State of Health (SoH) proxy and Remaining Useful Life (RUL) estimate
	Thermal and impedance anomaly score

Table 1. Measured versus estimated quantities. Only the left column is physically observed.

3.2 Why surface sensing is not enough

Heat generated in the cell core must conduct through the cell body to reach a surface sensor, so under changing load the core temperature can differ from the surface reading and the surface signal lags. A thermistor on the case therefore reports a delayed and attenuated view of the interior. A lumped thermal model driven by the measured surface temperature and the estimated heat generation can estimate the difference, but the estimate depends on thermal parameters (core-to-surface resistance and heat capacities) that must be identified. HeatSense treats T_{core} as an estimate whose accuracy has to be demonstrated, not as a measurement.

3.3 Hidden-state estimation with an ECM and an EKF

The 2RC equivalent-circuit model describes the terminal voltage as the open-circuit voltage minus an ohmic drop and two polarization voltages. Running this model in parallel with the cell yields a predicted terminal voltage. The difference between the predicted and the measured voltage (the innovation or residual) is used by the Extended Kalman Filter to correct the state estimate at every step. Because the residual measures disagreement between the model and the cell, it is also a natural signal for anomaly detection: sustained growth in the residual suggests that the cell is no longer behaving as modelled, for example because internal resistance has changed.

Internal resistance also links electrical and thermal behaviour. Ohmic heat is $Q = I^2R_s$, which is quadratic in current and linear in resistance. When R_s rises with ageing, the same current produces proportionally more heat, which is why tracking R_s matters for thermal monitoring. One source document states that the heat grows "exponentially" with resistance; that wording is incorrect for a fixed current and is not used here.

3.4 Physics-informed modelling compared with pure machine learning

A pure data-driven model maps inputs directly to outputs (for example V, I, T → SoH). It can be accurate inside the range of its training data, but it can behave unpredictably outside that range and gives little insight into why it produced a value. A physics-informed pipeline inserts a model between the raw data and the prediction, so the ML layer sees physically meaningful states. The expected benefits are better generalization and explainability. These are hypotheses for HeatSense to test, not conclusions, and the comparison in Section 7.5 is designed to test them.

Feature	Traditional BMS	BMS + Edge AI	BMS + Digital Twin + Edge AI
Primary purpose	Monitoring and protection	Prediction and forecasting	State understanding and simulation
Method	Threshold rules	Data-driven patterns	Physics-informed modelling plus data
Hidden-state estimation	Limited or none	Limited	Yes, model-based
What-if simulation	No	No	Possible with the model
Explainability	High (rules)	Low (black box)	Higher (explicit states)
Compute and complexity	Lowest	Moderate	Highest

Table 2. Architecture comparison adapted from the project sources. Safety-standard compliance is not claimed for any column.

3.5 Why the intelligence layer belongs at the edge

Protection must not depend on a network. A cloud round trip adds delay and fails when connectivity is lost, so any function that guards the cell has to run locally. One source illustrates the point with a five-second round trip, during which a vehicle at 80 km/h covers about 111 m. That figure is an illustrative scenario, not a measured latency, and HeatSense does not rely on it. The engineering conclusion that stands is simpler: the safety path stays local and deterministic, and the cloud, if ever used, is limited to non-safety analytics.

3.6 Hierarchical scaling and the average fallacy

A pack is limited by its weakest cell, and pack averages can hide it. As an illustration from the sources, a pack could report an average SoH of 93.5 percent while one cell sits near 82 percent. A cell twin, a module twin and a pack twin are the natural hierarchy for catching this. HeatSense implements only the cell twin. The module and pack levels appear in Section 12 as future work.

4. Technology Stack

Table 3 lists the planned technologies. All entries are choices made at the ideation stage; none has been implemented. Sampling rates and thresholds are design targets to be calibrated.

Layer	Technology	Role
Firmware	C++ / Arduino framework on Arduino UNO R4 Minima	Sensor sampling (INA219 at a 10 Hz target, DS18B20 at 1 Hz), fixed-threshold safety checks, relay driver, USB serial telemetry.
Sensing	INA219 (I^2C), 3× DS18B20 (1-Wire), optional BME280	Voltage, current, surface temperature, optional ambient reference.
Safety actuation	5 V relay (or MOSFET) module	Hard disconnect driven by firmware only.
Load	10 Ω ceramic power resistor	Controlled resistive load for low-rate discharge.
State estimation	Python, NumPy, SciPy on host laptop	2RC ECM, EKF and two-node thermal model.
Machine learning	Python, scikit-learn, PyTorch, XGBoost	Lightweight physics-informed regression and classification for SoH proxy and anomaly scoring; direct-ML baseline.
Backend	FastAPI (Python)	Serial ingestion, pipeline execution, API endpoints.
Dashboard	React web UI	Live measured values, estimated states, advisory alerts.
Offline datasets	NASA Ames, CALCE, Oxford, Stanford/MIT public battery data	Training and benchmarking before any bench data exists.
Optional interface	MCP2515 CAN module	Future automotive-bus interface; not part of the validated data path.

Table 3. Planned technology stack.

The project sources also mention an ESP32 and a Raspberry Pi as possible compute platforms, and one of them lists a physics-informed neural network (PINN). For this report the microcontroller is the Arduino UNO R4 Minima and the intelligence tier is the host laptop. The ESP32 and Raspberry Pi remain candidates for later migration to embedded inference. The ML layer is called “physics-informed” and not “PINN”, because a PINN implies an explicit physics-residual term in the training loss, which the project will adopt only if it is actually implemented.

5. System Approach

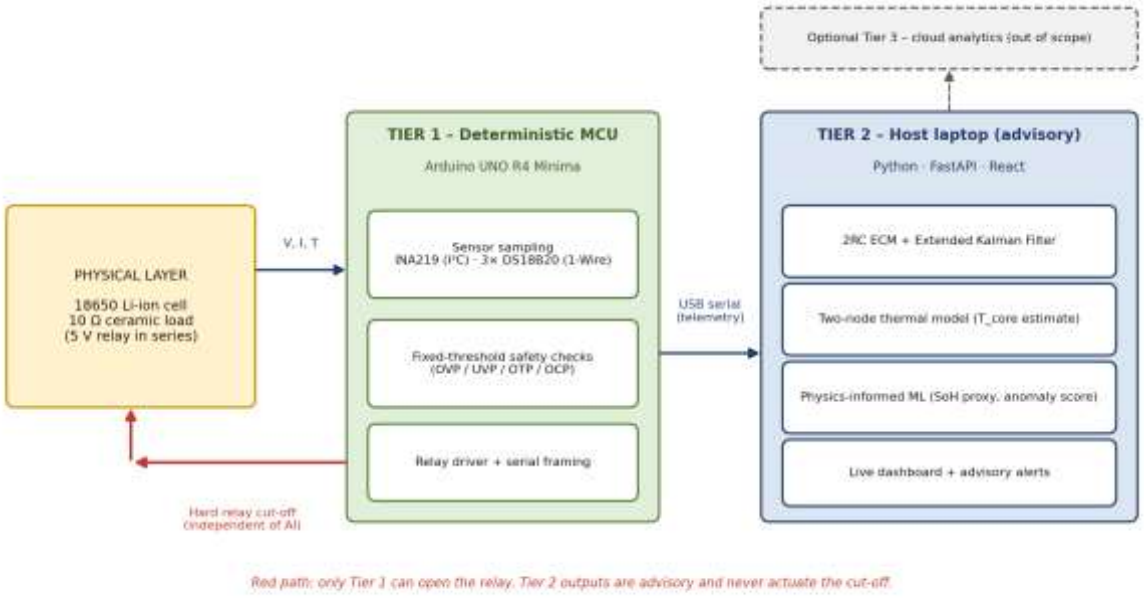


Figure 1. Planned architecture. The red path is the only route to the relay.

5.1 Tier responsibilities

Tier	Responsibilities	Must not do
Tier 1: deterministic MCU	Sample sensors; compare readings with fixed limits; open the relay on any violation; frame and send telemetry.	Depend on the laptop, the network or any ML output to make a safety decision.
Tier 2: host laptop	Ingest telemetry; run the ECM, EKF and thermal model; run ML; serve the dashboard; raise advisory alerts.	Command the relay or override Tier 1.
Tier 3: cloud (optional)	Historical storage, fleet analytics, offline retraining.	Appear in any safety path. Out of scope for the capstone.

Table 4. Division of responsibility between tiers.

5.2 End-to-end data flow

1. **Sensing.** The INA219 measures cell terminal voltage and load current over I²C; three DS18B20 probes measure surface temperature over 1-Wire.
2. **Deterministic safety check.** The microcontroller compares each reading with fixed limits. Illustrative defaults from the sources are over-voltage above 4.2 V, under-voltage below 2.5 V and surface over-temperature above 60 °C. These values must be calibrated to the actual cell datasheet before use.
3. **Relay action.** On any violation the microcontroller de-energizes the relay and isolates the cell. This step does not wait for the host.
4. **Telemetry.** If no limit is violated, the microcontroller packages the readings into structured serial frames and sends them over USB to the laptop.

5. **State estimation.** The host runs the ECM and EKF to estimate SoC, R_s, V_1 and V_2, and the thermal model to estimate T_{core}.
6. **Advisory ML.** The estimated state vector and the EKF residuals feed the ML layer, which outputs a SoH proxy and an anomaly score.
7. **Visualization.** Measured values, estimated states and advisory outputs are shown on the dashboard.

5.3 Communication choice

The primary link from the microcontroller to the laptop is USB serial. The MCP2515 CAN controller is listed as optional hardware for demonstrating CAN message framing, but unless a USB-CAN adapter is added on the host, CAN is not in the data path and the report makes no claim about it.

5.4 Safety separation rules

- The relay is driven only by firmware on Tier 1.
- A fault in Tier 2 (crash, wrong estimate, lost connection) cannot prevent a Tier 1 cut-off.
- Dashboard status “Normal” means that no configured bench threshold has been violated. It is not a statement that the battery is healthy or safe.
- Tier 2 can display warnings, but the response to a warning is a human decision.

6. Mathematical Formulation

This section defines the models the project intends to implement. Parameter values are not yet identified, and no equation has been validated against a physical cell.

6.1 State and measurement variables

$$x = [\, SoC,\ V_1,\ V_2,\ R_s,\ T_{core} \,]^T$$

Inputs are the measured current I and ambient temperature T_{amb}. Outputs are the measured terminal voltage V_t and surface temperature T_{surf}. SoC, V_1, V_2, R_s and T_{core} are never measured and must be estimated.

6.2 Second-order Thevenin equivalent circuit

$$V_t = V_{OCV}(SoC) - V_1 - V_2 - I\,R_s$$

V_{OCV} is the open-circuit voltage as a function of SoC; V_1 models fast charge-transfer polarization with time constant $\tau_1 = R_1C_1$, and V_2 models slower diffusion with $\tau_2 = R_2C_2$. In discrete time with step Δt:

$$V_1(k+1) = e^{-\Delta t/\tau_1}\, V_1(k) + R_1\,(1 - e^{-\Delta t/\tau_1})\, I(k)$$
$$V_2(k+1) = e^{-\Delta t/\tau_2}\, V_2(k) + R_2\,(1 - e^{-\Delta t/\tau_2})\, I(k)$$
$$SoC(k+1) = SoC(k) - (\,\eta\,\Delta t\,/\,(3600\,Q_n)\,)\, I(k)$$

Here η is coulombic efficiency and Q_n is the cell capacity in ampere-hours. The sign convention for I must be fixed once in firmware and kept throughout.

6.3 Extended Kalman Filter

The filter tracks x = [SoC, V_1, V_2]. Each step has a prediction and a correction. The EKF is used as a recursive model-based estimator for a nonlinear system; it is not claimed to be statistically optimal.

$$\hat{x}_{k|k-1} = f(\hat{x}_{k-1|k-1}, I_{k-1}), \quad P_{k|k-1} = A\,P_{k-1|k-1}\,A^T + Q$$

$$K_k = P_{k|k-1}\,C^T\,(C\,P_{k|k-1}\,C^T + R_{meas})^{-1}$$

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k\,(V_{meas} - V_{pred}), \quad P_{k|k} = (I - K_k\,C)\,P_{k|k-1}$$

A is the Jacobian of the state update (diagonal here, with entries 1, e^{−Δt/τ_1}, e^{−Δt/τ_2}) and C = [dV_{OCV}/dSoC, −1, −1]. Because a three-state filter has a 3×3 covariance matrix, the numerical load is small. The reason to run it on the host first is convenience for development and parameter identification; whether it can later run on the microcontroller is a measurement to be made, not an assumption.

6.4 Resistance tracking on two time scales

R_s changes slowly with ageing and temperature, while SoC, V_1 and V_2 change every step. The planned approach is therefore two-timescale: the fast EKF runs every sample with fixed parameters, and a slower routine re-identifies R_s, R_1, C_1, R_2 and C_2 from pulse-discharge segments (for example a short on/off load profile, where $R_s \approx \Delta V/\Delta I$ at the step). The resistance-based SoH proxy is computed from the slow estimate of R_s relative to a reference value.

6.5 Two-node thermal model

$$C_{core}\,dT_{core}/dt = Q_{gen} - (T_{core} - T_{surf})/R_{cond}$$

$$C_{surf}\,dT_{surf}/dt = (T_{core} - T_{surf})/R_{cond} - (T_{surf} - T_{amb})/R_{conv}$$

$$Q_{gen} = I^2\,R_s + I\,T_{core}\,(\partial V_{OCV}/\partial T)$$

The first term of Q_{gen} is ohmic heating and the second is the reversible entropic term. C_{core} and C_{surf} are heat capacities, R_{cond} is the core-to-surface thermal resistance and R_{conv} is the surface-to-ambient resistance. These parameters are initially taken from literature or datasheets and must be refined from experiments; until then T_{core} is an uncalibrated estimate.

7. End-to-End Methodology

7.1 Sensing and pre-processing

Voltage and current are sampled at a target of 10 Hz and temperature at 1 Hz, because the DS18B20 converts slowly and the thermal time constant of a cell is long. Readings are time-stamped by the microcontroller and framed with a checksum so that corrupted frames are dropped rather than passed to the estimator. Sensor noise is handled by the EKF measurement-noise term; extra filtering is added only if the data show it is needed.

7.2 State estimation

The host runs the ECM and EKF on each frame, then updates the thermal model with the latest surface temperature and the estimated R_s. The outputs are the state vector and the innovation sequence. An initial SoC is taken from an open-circuit-voltage lookup after rest, and the OCV–SoC curve is to be measured for the actual cell with a slow charge and discharge instead of being taken from a generic table.

7.3 Anomaly detection from EKF innovation

The innovation $e_k = V_{meas} - V_{pred}$ has a predicted variance $S_k = C\ P\ C^T + R_{meas}$. Normalizing the innovation by S_k gives a quantity whose size can be monitored. Persistent growth of the normalized innovation, a rising R_s estimate, or an abnormal rate of change of surface temperature under a given load are treated as impedance or thermal precursor indicators.

This gives a first, physics-only anomaly flag that needs no training data. The alarm threshold is chosen from the innovation statistics measured during normal operation, and the false-alarm rate is a metric to be reported. The flag is advisory; it does not open the relay.

7.4 Machine-learning prognostics

The ML layer receives a rolling window of estimated states and residual trends (SoC, R_s, V_1, V_2, T_{core} estimate, innovation statistics, cycle count) and outputs a SoH proxy, an RUL estimate if a failure criterion is defined, and an anomaly score. Candidate models are a compact LSTM or gradient-boosted trees; the final choice follows the benchmark.

- Training data: public datasets (NASA Ames, CALCE, Oxford, Stanford/MIT) are used for offline training and benchmarking.
- Physics consistency: a soft penalty can discourage physically implausible outputs, such as SoH rising without intervention. A soft penalty encourages plausible behaviour; it does not guarantee it.
- Labelling: SoH proxy values derived from resistance are labelled as proxies. RUL is reported only if an end-of-life criterion (for example capacity below a stated fraction of nominal) and a degradation trajectory are defined; otherwise it is reported as “not validated”.
- Transfer risk: a model trained on public cells may not transfer to the bench cell, and the report must say so when results are given.

7.5 Validation: the central experiment

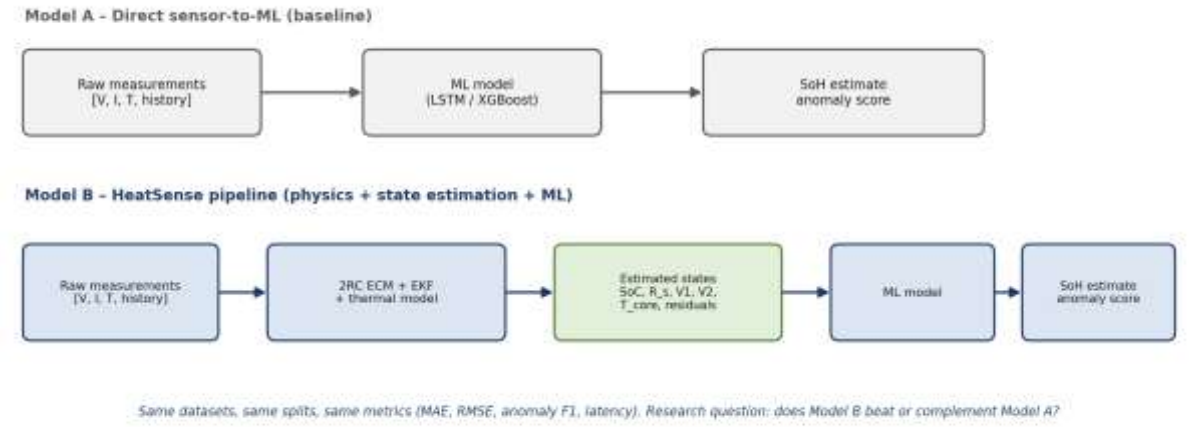


Figure 2. Model A (baseline) and Model B (HeatSense) are trained and tested on identical data.

Model A maps raw measurements directly to the targets. Model B first passes the same measurements through the ECM, EKF and thermal model and gives the ML layer the estimated states. Both are trained on the same datasets with the same splits and are compared on the same held-out cells. The

hypothesis is that Model B gives lower SoH error, a higher anomaly F1 score, better behaviour on operating conditions unseen in training, or comparable accuracy with better interpretability.

Metric	Model A (direct ML)	Model B (HeatSense)
SoH MAE	[not yet measured]	[not yet measured]
SoH RMSE	[not yet measured]	[not yet measured]
Anomaly detection F1	[not yet measured]	[not yet measured]
False-positive rate on normal data	[not yet measured]	[not yet measured]
Generalization to unseen conditions	[not yet measured]	[not yet measured]
Inference latency per sample	[not yet measured]	[not yet measured]
Memory footprint	[not yet measured]	[not yet measured]

Table 5. Evaluation metrics. No value exists yet.

Decision rule

The added complexity of Model B is justified only if it achieves a lower SoH MAE or a higher anomaly F1 than Model A on held-out data, or matches Model A while giving clearer explanations. If it does not, the report states this outcome.

Additional experiments

- Software-injected faults: an artificial step in R_s, an artificial drift in a temperature channel, and an abnormal temperature-versus-load trend are added to logged data to test the anomaly flag.
- Safety-path test: a deliberately set threshold violation must open the relay within a measured time, recorded with a scope or logic analyzer, with the laptop disconnected.
- Timing: execution time of the firmware loop and of each host stage is measured, not assumed.

8. Planned Hardware

> **Status**
> No hardware has been bought, assembled or tested. This section describes the intended bench setup only.

Component	Planned role	Interface
18650 Li-ion cell (3.7 V nominal)	Single test cell in a spring-contact holder.	Direct wiring
Arduino UNO R4 Minima	Tier 1 controller: acquisition, thresholds, relay drive, serial framing.	I^2C, 1-Wire, GPIO, USB
INA219 module	Terminal voltage and load current.	I^2C
DS18B20 × 3	Surface temperature at three points on the cell.	1-Wire with pull-up
5 V relay (or MOSFET) module	Hard disconnect on fault.	GPIO via driver
10 Ω ceramic power resistor	Resistive load for low-rate discharge.	In series with relay
MCP2515 CAN module (optional)	Future automotive-bus demonstration.	SPI

Component	Planned role	Interface
BME280 (optional)	Ambient temperature reference for the thermal model.	I²C
Host laptop	Tier 2: estimation, ML, dashboard.	USB serial

Table 6. Planned bench hardware.

8.1 Why one cell

The research question concerns the estimator, and an estimator is validated first on a single cell under controlled conditions, as battery-modelling practice does before moving to packs. A series pack would also raise the safety risk and would need per-cell voltage sensing and balancing, which a single pack-level current and voltage sensor cannot provide. Multi-cell work is therefore deferred. One useful property of the single-cell design is that three surface probes at different positions can show a temperature gradient across the cell body.

8.2 Bench safety notes

- Discharge at a low rate through the 10 Ω load; the project does not charge, overcharge or abuse the cell.
- Keep a fuse or equivalent current limit in the circuit independent of firmware.
- Test unattended only if the relay path has been verified. Do not run destructive tests.

9. Implementation Roadmap

The plan follows the three phases in the project mind map. All items are planned; none has started. Phase boundaries are defined by exit criteria that can be checked, so progress can be reported without relying on impressions.

Phase	Planned activities	Deliverable	Exit criterion
1. Hardware and data	Procure parts; wire the bench; write acquisition firmware; implement fixed-threshold relay logic; define the serial frame; log raw data.	Working bench that streams V, I, T to the laptop; safety path verified without the laptop.	Relay trip time measured; logged data are clean and time-stamped.
2. Modelling and estimation	Measure the OCV–SoC curve; identify ECM parameters from pulse tests; implement the EKF and thermal model; implement the innovation-based anomaly flag.	Estimator running on logged and live data.	Voltage prediction error and SoC error against a coulomb-count reference reported.
3. Intelligence and actions	Train Model A and Model B on public datasets; run injected-fault tests; build the dashboard and alerts; measure latency and memory.	Benchmark report; live dashboard.	Table 5 filled with measured values; decision rule applied.

Table 7. Planned phases. Status: not started.

9.1 Planned bench demonstration

Once the phases are complete, a short repeatable demonstration is planned: normal low-rate discharge with live estimated states; a software-injected anomaly that raises the anomaly score; and a forced threshold violation that opens the relay with the laptop disconnected. The last step is the main evidence that the AI layer is not in the safety path.

10. Risks and Mitigations

Risk	Effect	Mitigation
Public-dataset cells differ from the bench cell	Model B may not transfer; reported accuracy may overstate bench performance.	Report results per dataset; fine-tune or recalibrate on bench data; state the transfer gap.
ECM parameters drift with temperature and ageing	Residuals rise without a real fault.	Two-timescale identification; log temperature with every estimate; review the residual threshold on normal data.
Thermal parameters taken from literature	T_{core} estimate may be biased.	Treat T_{core} as uncalibrated; calibrate against a controlled temperature step where possible.
Sensor noise or a loose connection	False alarms.	Frame checksums; plausibility checks; compare the three DS18B20 channels.
Scope creep (packs, Transformers, cloud)	Several unfinished projects.	Keep to one cell and the A-versus-B experiment; move the rest to future work.
Over-claiming results	Loss of credibility.	Label estimates; report "[not yet measured]" until measured; keep the exclusions in Section 11.
Low-rate bench cannot show degradation	SoH trend cannot be observed on a single cell in the project time.	Use public aging data for health benchmarks; use the bench for state estimation and fault injection.

Table 8. Risk register.

11. Limitations and Out of Scope

11.1 Limitations

- Single cell: pack behaviour, cell-to-cell imbalance and thermal propagation are not addressed.
- Core temperature is estimated by a lumped model and is not measured; its accuracy depends on parameters that still need identification.
- The 2RC ECM is a lumped approximation. It does not represent concentration gradients, particle cracking or SEI chemistry.
- SoH from resistance is a proxy. It is not equal to capacity-based SoH and is not validated against a capacity test.
- RUL requires a failure criterion and a degradation trajectory. Neither is available from a short bench test.
- No destructive testing is planned, so thermal runaway cannot be validated and is not claimed.

- The prototype makes no claim of compliance with automotive functional-safety standards such as ISO 26262.

11.2 Claims HeatSense does not make

Not claimed	Used instead
Thermal-runaway prediction or prevention	Thermal and impedance precursor anomaly detection
Measured SoH, measured RUL, measured core temperature	Estimated SoH proxy, RUL (only if validated), estimated T_{core}
"Normal" means safe	"Normal" means no configured bench threshold was violated
Validated EV-pack or hierarchical twin	Cell-level prototype with a scalable architecture
Automotive-grade Edge AI deployment	Laptop-based analytics tier for the prototype
Transformer or foundation-model prognostics	Compact physics-informed model; sequence models as future research
Standards compliance	No compliance claim

Table 9. Wording the report avoids and the wording it uses.

12. Future Work

- **Multi-cell module twin.** Extend to two cells in series with a voltage tap per cell (for example one INA219 per cell at different I^2C addresses), then to a module, to study imbalance and weak-cell identification. Pack-level balancing needs a proper multi-cell monitor and is a separate project.
- **Quantized embedded inference.** Port the EKF and a compact model to fixed-point code and measure execution time and memory on a microcontroller or a small single-board computer; this is the point at which an "edge AI" claim becomes measurable.
- **Battery-state sequence models.** Treat sequences of battery state vectors as inputs for next-state prediction, for example with a Transformer or a state-space model, $P(X(t+1) \mid X(t), X(t-1), \ldots, X(t-k))$. This is research that should start only after the Model A versus Model B result exists.
- **Native CAN link.** Add a USB-CAN adapter on the host so that telemetry can be demonstrated over CAN.
- **Cloud analytics.** Fleet-level storage and retraining, kept outside any safety path.

13. Source Notes

This report was prepared only from the three project documents supplied to the writing tool. No external citations are added.

#	Source	Used for
S1	HeatSense idea report generated in NotebookLM (problem, solution, stack, approach, methodology, limitations, future work).	Structure of sections 1–5, 7, 11, 12.
S2	Technical research report “The Critical Necessity of Digital Twins in EV Battery Management”.	Background in section 3 (observable and latent states, physics-informed versus pure ML, architecture comparison, hierarchy).
S3	HeatSense mind map (three-tier architecture, hardware list, Digital Twin versus direct ML, three implementation phases).	Hardware list, phases in section 9, optional sensors.

Table 10. Supplied sources.

The datasets named in this report (NASA Ames, CALCE, Oxford, Stanford/MIT) are named in the sources but their files and papers were not supplied, so nothing is quoted from them. Figures such as the 40 to 60 percent battery cost share, the five-second latency scenario and the 93.5 versus 82 percent SoH example are reported as they appear in the sources and have not been verified independently.

13.1 Source statements not carried into this report

Statement in a source	Reason it is excluded or changed
Heat grows “exponentially” as resistance rises	With $Q = I^2R$ at fixed current, heat is linear in R and quadratic in I.
“Predicting failure before damage happens” and thermal-runaway prevention	Cannot be validated on a low-rate, non-destructive bench; replaced by precursor anomaly detection.
Field case of one cell with 42 voltage dips in seven days	No data or reference supplied; treated as an illustration of intermittent-fault detection only.
Architecture 3 “ISO/TR 9968:2023 compliance”	No compliance is claimed.
PINN, ESP32 and Raspberry Pi in the mind map	Replaced by “physics-informed ML”, Arduino UNO R4 Minima and a host laptop for this report.
Three tiers as a core design	Cloud tier is optional and out of scope, so the working design has two tiers.

Table 11. Conflicts between sources and the locked decisions.

Appendix A. Glossary

Term	Meaning
BMS	Battery management system: monitoring and protection electronics for a battery.
Digital Twin (here)	A real-time state estimator that keeps a parametric model synchronized with one physical cell.
ECM / 2RC	Equivalent-circuit model with a series resistance and two RC polarization branches.

Term	Meaning
EKF	Extended Kalman Filter: recursive model-based state estimator for nonlinear systems.
SoC	State of charge: fraction of capacity remaining (estimated).
SoH (proxy)	State of health. Here a resistance-based proxy until checked against capacity.
RUL	Remaining useful life: cycles left before a defined end-of-life criterion.
T_{core}	Estimated cell core temperature from the two-node thermal model.